LoST: Level of Semantics Tokenization for 3D Shapes

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Research Questions

- How can 3D shapes be tokenized by semantic salience rather than geometric level-of-detail to enable efficient autoregressive generation?
- How can semantic supervision be provided for 3D shape tokenization without direct access to semantic labels?
- Can semantically-ordered tokens enable high-quality 3D generation with dramatically fewer tokens than geometric hierarchies?

Core Idea

Current 3D tokenization methods use geometric level-of-detail (LoD) hierarchies that require many tokens to represent basic shapes and produce unusable early prefixes. This paper proposes ordering tokens by semantic salience instead - early tokens capture the core semantic identity (e.g., 'chair') while later tokens add instance-specific details. The key insight is using Relational Inter-Distance Alignment (RIDA) to train a semantic extractor that aligns 3D shape relationships with DINO image feature relationships, providing semantic supervision without needing explicit 3D semantic labels.

Key Formula

Inter-Instance Rank Distillation Loss

$$L_{rank} := E_{z_i \in B}\left[|| \tilde{c}_i^s - \tilde{c}_i^t ||_2^2 \right]$$

Aligns relative similarity rankings between student (3D) and teacher (DINO) spaces. \tilde{c}_i^s and \tilde{c}_i^t are z-scored cosine similarity vectors that remove modality-specific means/scales, making the loss invariant to absolute similarity values while preserving neighborhood rankings.

RIDA Combined Loss

$$L_{RIDA} := \lambda_g L_{global} + \lambda_r L_{rank} + \lambda_s L_{spatial}$$

Total semantic alignment objective combining global contrastive loss (L_global), inter-instance rank distillation (L_rank), and spatial structure distillation (L_spatial). Used to train the semantic extractor that maps 3D triplanes to semantically-structured features aligned with DINO space.

Key Figure

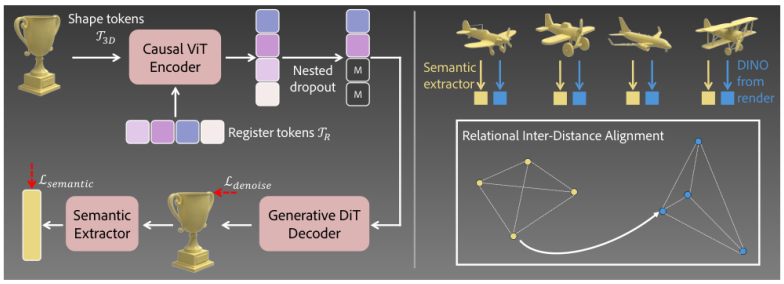

Figure 2. **Overview of LoST**. Left: LoST maps 3D shape latents into a token sequence ordered by semantic salience, where early prefixes capture coarse semantics and later tokens refine instance-specific detail. A conditional generative DiT decoder reconstructs the complete latent from any prefix. Right: The semantic extractor is pretrained with Relational Inter-Distance Alignment (RIDA), which aligns relationships in 3D latent space with DINO feature relationships to provide semantics-aware supervision.

high-frequency details, analogous to spectral analysis in 3D geometry [31], and a semantic loss gives us a more semantic hierarchy. We use 768 triplane tokens after patchification and a maximum of $k = 512$ register tokens $\mathcal{T}_R$ as we see marginal improvement beyond this.

3.2. LoST Decoder

We aim to decode the full sequence of 3D latents from any prefix of the register tokens. However, reconstructing the complete geometric signal from very few tokens is inherently challenging, as the ambiguity inherent in the limited information results in blurry, coarse reconstructions when decoded deterministically. Instead of exact geometric reconstruction from very few tokens, we focus on producing *semantically plausible* reconstructions that may differ in geometry. To this end, following [1, 35], we reframe the task as a generative problem and employ a diffusion model to produce the full sequence conditioned on a variable-length prefix of the encoded register tokens. As the prefix length increases, generation gradually transitions toward reconstruction, since longer prefixes reduce ambiguity in the predicted sequence. More concretely, we train a Diffusion-Transformer (DiT) model [13, 24], $\mathcal{G}$ to reproduce the full signal conditioned on a flexible prefix of $\mathcal{T}_R$, which is obtained by simply masking out the unused postfix. The generator $\mathcal{G}$ takes as input noisy shape tokens and predicts the added noise by cross-attending to the conditional $\mathcal{T}_R$. See supplemental for details.

3.3. Semantic Guidance for Learning LoS Tokens

To improve the semantic structure, both FlexTok and Semanticist [1, 35] have relied on Representation Alignment [41] (REPA) loss to enforce alignment between internal representations of the diffusion model and semantic DINO [16] features extracted from the target image. The inclusion of such a DINO-based semantic REPA loss encourages the tokens to encode semantics and enables the learned hierarchy to capture progressively richer levels of semantics within the token sequence. However, no comparable semantic feature extractor and alignment loss exist for 3D shape generation, making it challenging to directly apply REPA-style supervision in our setting. Directly aligning the internal representations of our 3D generative model with those of a 2D visual foundation model (*e.g.*, DINO) performs poorly, even when reconstructing from the complete set of register tokens. The failure can be attributed to differences in the spatial layout and inherent dimensionality of the two representations. An option to align the dimensionality of the two representations could be to apply the REPA loss to multi-view renders of decoded triplanes, but this is computationally prohibitive.

Relational Inter-Distance Alignment (RIDA). Our key insight is that we only need to align *contrastive relative distances* in corresponding sample sets of the two representations, rather than regressing absolute values. Therefore, we define a mapping from the triplane latent space into a new feature space where relative distances match those of DINO. Once this mapping is established, we can use this feature space instead of DINO for semantic guidance. Specifically, we propose *Relational Inter-Distance Alignment (RIDA)*, a novel pre-training for creating this mapping to a *student* feature space, where relative distances are aligned to a *teacher*.

In our setting, the teacher space is formed by DINO features. As our training set consists of 3D shapes reconstructed from generated images, we encode DINO features directly from the generated images, giving us spatial tokens $\mathbf{S}_i^t \in \mathbb{R}^{K \times d}$ and a further global embedding $\mathbf{z}_i^t \in \mathbb{R}^D$. To

Figure 2: Shows the complete LoST architecture with causal ViT encoder, nested dropout, generative DiT decoder, and RIDA semantic alignment training pipeline.

Limitations

- Relies on VAE triplane representation, limiting extension to other 3D formats without adaptation
- Uses diffusion decoder which increases computational cost compared to pure autoregressive decoding
- Few-token regime still exhibits artifacts compared to full-length decodes
- AR generator uses fixed target length rather than variable-length with EOS tokens for complexity-aware stopping